



Siamese BERT Architecture Model with attention mechanism for Textual Semantic Similarity

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Received: 12 November 2021 / Revised: 6 June 2022 / Accepted: 19 April 2023 /

Published online: 2 May 2023

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Abstract

Textual Semantic Similarity is a crucial part of text matching tasks, and it has a very wide range of applications in natural language processing (NLP) tasks such as search engines, question-answering systems, information retrieval, natural language inference. Although there are a variety of approaches about textual semantic similarity, many do not succeed in achieving the semantic representation of a sentence or text that represents it well, and ignore that different words serve different roles in expressing the meaning of the whole sentence in different degrees. Therefore, our paper proposes a Siamese Bert network model to obtain textual semantic similarity. Firstly, we utilize the Bert network model to obtain the semantic features of each word in the sentence as input and utilize the merit of the Siamese network, reducing the training parameters, sharing the same encoder and feature weight information with each other. Then we use the attention mechanism to obtain more advanced semantic features. Furthermore, the similarity between two sentences can be derived by the methods of calculating the distance or concatenating their high-level semantic representations. In this paper, we apply the network structure to three related semantic similarity datasets, which perform better than other approaches.

Keywords Siamese network · Bert · Attention mechanism · Textual semantic similarity

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1 Introduction

Textual semantic similarity purposes to judge the similarity between two sentences or two texts, which makes it very significant and widely used in the field of NLP. For instance, in the natural language inference task, we often need to judge the semantic relationship between two sentences through the textual semantic similarity. In the question answering (QA) system, we will match some questions and answers which have a strong correlation with each other and also means a high degree of similarity. In the information retrieval, we would obtain the most relevant text information with our target information, and the two also have a high degree of similarity. Therefore, how to calculate textual semantic similarity becomes an indispensable task for the field of NLP.

Previously, similarity was obtained primarily by keyword statistics and matching, which relied on mathematical statistical methods including N-grams and TF-IDFs. Now with the rapid development of deep learning, using deep neural network models to solve feature expression also have many applications and good performance in this task. For instance, convolutional neural network (CNN) model and long short-term memory (LSTM) model have been proposed to calculate textual similarity. Using these neural network models obtains high-level semantic representations of sentences or texts, and calculates the distance between the semantic representation vectors obtained from different sentences or texts to measure the similarity between them. In addition to these serialization models, researchers found the framework of Siamese network model has better performance in evaluating textual semantic similarity tasks so they gradually combined this Siamese network model with the specific network model to implement semantic similarity tasks. For example, researchers evaluated the semantic similarity via a Siamese adaptation of the LSTM network model [27] and proposed the Siamese LSTM model for evaluating the semantic similarity in asymmetric text [45]. If we change the two branches of shared parameters in the Siamese network structure to two independent branches, we will get the Pseudo-Siamese network. The two branches of Pseudo-Siamese network can have the same structure but different weights, or can be a completely different structure. Pseudo-Siamese network mostly were used in the field of computer version such as [40].

The latest achievement of the pre-trained model Bert [7] has widely applied to various NLP tasks. The Bert model used Transformer as the feature extractor, it can deeply comprehend context information and better express the complexity of words. Moreover, the Bert model can train the weights of the classifier and fine-tune the pre-trained layer to allow terms from different domains to be further adapted for use in various tasks. In our paper, we use Bert, a general model, for text feature extraction, apply the method to text similarity tasks, and train on datasets from different domains. The Bert model has strong adaptability when applied to special word features in different fields, and can be fine-tuned according to special words in different fields, so that it has different semantic representations in different fields.

It is possible to preserve semantic representation and improve the prediction performance of similarity between sentences using a combination of neural network models. Nevertheless, these methods neglected different words have various donation to the semantic expression of the entire sentence. Therefore, we have to propose the concept of attention mechanism. The attention mechanism was originally applied to computer vision. In the field of computer version, researchers propose a progressive multi-scale feature extraction block to extract feature maps of different sizes. Furthermore, the HR-size expectation-maximization attention block aims to enable a balance between performance and applicability

by directly capturing the long-range dependencies of HR-size feature maps [47]. A cross view capture network is employed in this analysis of stereo image super-resolution in order for global and local contextual features to be taken into account [46]. The introduction of the attention mechanism enables our model to evaluate and identify which words encapsulating semantic features in a sentence are of more importance and to give these words a different weight.

Based on the background of this task, we propose a Siamese Bert network model structure with the attention mechanism in our paper to predict the similarity among two sentences and apply it to the datasets of three related semantic similarity tasks. Experimental results on all datasets show that our methods achieve better performance than the baseline models and traditional methods. The main contributions of our paper are as follows:

- We utilize Bert model to encode sentences and generate the meaningful semantic feature-vector in the pre-training process.
- It is proposed to utilize the Siamese network based Bi-LSTM model to predict the textual semantic similarity and use the characteristics of the Siamese network to share weights, reduce the complexity of the model, and prevent too easy to overfitting during the training process.
- Our model with attention mechanism assigns different weights to each word embedding in the sentence to enhance the semantic features of the sentence and obtain the high-level semantic representation for better to distinguish sentence differences and adapt to semantic similarity tasks.
- To evaluate the similarity predictions of our model with respect to various metrics such as accuracy and F1-score value, we validated it on three similarity-related task datasets in the field of NLP and demonstrate the generality of our model related to semantic similarity task.

The rest of the paper is organized as follows. Section 2 introduces a review of related works comprising textual semantic similarity and the recent methods. Section 3 describes our architecture model and the methods using in each layer. Section 4 shows our experimental setup. Section 5 depicts our experimentations and results. Section 6 is a conclusion about our model method and experimentation and future research directions and ideas for this paper.

2 Related work

In the field of NLP, calculating similarity between texts and sentences has become an essential task. The task of textual semantic similarity is simply to evaluate the similarity between two texts or two sentences. Therefore, there were many applications such as information retrieval, machine translation [41], QA systems [16], and so on. It is not uncommon for two sentences to have two different expressions between them, but their semantics are the same and how to aid a computer in distinguishing these situations is a difficult task for text semantic similarity. Researchers had also proposed many methods to solve how to gain the better semantic representation.

The Bert model has achieved new state-of-art performance in the various NLP tasks such as QA system, text classification and sequence label problems. Bert model was called Bidirectional Encoder Representations from Transformer [7] which has the

ability to seize self-attention through the Transformer structure. The Bert model utilized Transformer as the feature extractor, it can make good use of context information. Recently, the Bert model as a pre-trained model has worked on much data from specialized fields. A domain-specific language representation model trained on large biomedical corpora is BioBERT (Bidirectional Encoder Representations from Transformers for Biomedical Text Mining) in the field of biomedicine [18]. Based on bidirectional transformers (ClinicalBert), this work develops and evaluates clinical notes representations. ClinicalBert uncovers high-quality relationships between medical concepts as judged by humans [13]. Besides, the Bert network used Siamese network structure to catch contextual semantic and generate the meaningful sentence embeddings that can be compared using cosine-similarity [32]. So Bert model as pre-trained model can improve whole model performance. Apart from that, the Bert model was originally trained on general corpus and had strong adaptability when applied to special word features in different fields, and can be fine-tuned according to special words in different fields. Researchers proposed a topic-informed based on Bert architecture for pairwise semantic similarity detection [29]. The Bert model is proven to be successful in qualitative analyses using domain-specific words.

The semantic representation of the text transforms the words and semantics of a sentence into digital features that are directly calculable. In the beginning, researchers used various features of text for paraphrase identification tasks including n-gram features [24], linguistic features [34], etc. Then, machine learning method were used for this task such as SVM [35]. As a result of the successful use of neural networks in major NLP tasks, researchers were investigating the use of neural network models for text semantic representation. Typically, the CNN model and the RNN model performed in learning the sentence representations and the relationship among different elements in the input sentence. Researchers utilized the CNN model to solve textual semantic similarity task [36]. A semantic similarity score between two sentences was calculated by comparing their semantic vectors. Modeling sentence similarity was complicated by the ambiguity and variability of linguistic expression, so this research proposed a model for comparing sentences that used a multiplicity of perspectives [12]. An approach based on convolutional neural networks first models each sentence using features extracted at multiple levels of granularity and pooled in multiple ways. Nowadays, many methods aimed to gain a better understanding of the text's context in order to better express its semantics. It was then discovered that the Siamese network architecture can be used to evaluate semantic similarity, since it has the ability to encode sentence pairs using equal weights and parameters and to produce similar semantic sentence representations using the same parameters. Researchers combined convolution and recurrent neural networks to measure the semantic similarity of sentences [30]. In this study, a convolution network was used to account for the local context of words, and an LSTM was used to account for the global context of sentences. The combination of these networks helped to preserve the relevant information of sentences and improve the calculation of similarity between them. Siamese network model was a framework of a neural network, not a specific neural network model. Therefore, the LSTM model, CNN model, or gate recurrent unit (GRU) model also can apply to its implementation. This study examined the problem of normalizing job title sequences using a deeper construction for learning a similarity metric on variable-length character sequences [28]. Combining a stack of bidirectional LSTMs at the character-level with a Siamese architecture, this algorithm projected variable-length strings into a fixed-dimensional embedding space by examining their similarities. Hence, it is realistic to apply the Siamese network to the task of semantic similarity.

Although this method concentrated on the semantic features of words or a single sentence, different words play a role in the semantic expression of the entire sentence to varying degrees. Therefore, the attention mechanism was adopted to endow different weights to each word in the sentence and had the wide application on many NLP tasks. An attention-fused deep matching network (AF-DMN) was used to fully exploit local context information within each sentence for the natural language inference task [9]. AF-DMN took two sentences as input and iteratively learned the attention-aware representations for each side by multi-level interactions. For words distance, a distance-based Self-Attention model was proposed, which considers the word distance by using a simple distance mask, because it will not lose the ability to model the global dependency inherent in attention but also model the local dependency [14]. For biomedical sentence similarity task, researchers utilized the attention mechanism to obtain the weights of words in a single sentence and enhance sentence semantic representation on the biomedical dataset [19]. Likewise, attention constituency vector tree (ACVT) was proposed for semantic similarity task [31]. A new tree kernel was developed, known as ACVT kernel, which is customized for measures of sentence similarity based on the proposed structure. For determining how self-attention scores encode semantic content, researchers applied the LAT probing method in the Transformer to trace attention over multiple layers for sentiment analysis tasks [44]. In addition to sequence labeling, the attention model is used for learning attention segmental recurrent neural networks based on hierarchical attention neural semi-Markov conditional random fields [20]. To enhance the computation efficiency and performance, researchers introduce syntactical and semantical attention, which assign different weights to different parts of the input sentence [22]. End-to-end self-attention-based Conditional Random Fields can be used for automatic feature extraction and selection of best encoding schema for sequence labeling [37]. A bidirectional tree-structured LSTM with an attention mechanism has been developed to extract positional information of entity pairs and to determine what words comprise the semantic relation between the entities [10]. The attention mechanism is also utilized in the domain of Internet-Delivered Psychological Treatment. Research has investigated the application of personalized mental health interventions using bidirectional LSTM architecture with attention mechanism [2]. As a result of the attention mechanism, the model is capable of handling tasks from various domains and has greater expressive abilities. Hence, our paper introduced the attention mechanism to obtain weights of words and strengthen the semantic representation of sentences in our model.

3 Methodology

In this section, we will describe how the Siamese Bert architecture model solves textual semantic similarity tasks. Figure 1 shows the Siamese Bert architecture and the model processing details. Our model has four layers including input layer, Siamese feature extract layer, attention layer, predicted layer. Firstly, we explain the Bert model. Then, we introduce our Siamese structure network based on Bi-LSTM model for extracting semantic features. Moreover, we utilize Attention Mechanism to obtain higher-level semantic representation. Finally, the method of prediction similarity and the loss function used in our experiments will be introduced.

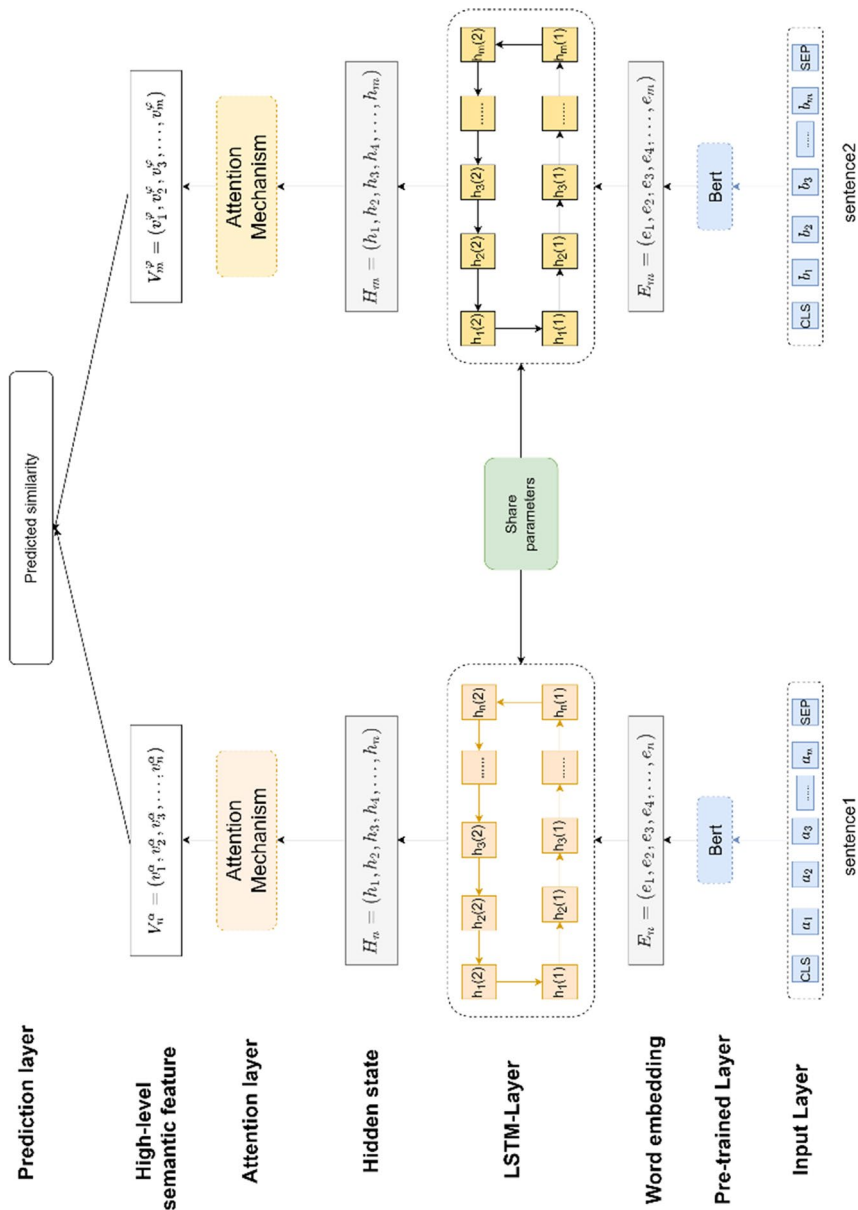


Fig. 1 The model architecture

3.1 Pre-trained model of Bert model

In the past, the first-generation pre-training model was mainly to obtain a static word vector representation such as Glove, Word2Vec. Nowadays, the second-generation pre-training model mainly obtains context-related word vector representations including Bert. Bert model is the Bidirectional Encoder Representation from Transformer. The input of Bert consists of Token Embedding, Segment Embedding and Position Embedding. The '[CLS]' is the beginning mark of the entire sentence. The '[SEP]' is the mark which separates two sentences. We show a sentence of “Children are smiling and waving at camera” architecture as an input example in Fig. 2.

The pre-training phase of Bert includes two tasks, one is Masked Language Model (MLM), and the other is Next Sentence Prediction (NSP). The MLM task can be understood as cloze filling. Each sentence as input will be randomly masked by 15% of the words, and then using the method of unsupervised learning predicts the word in the mask position. But at the same time, in order to prevent that some words in the mask 15% of the words have never been seen in the fine-tuning stage, the model process the sentence in a different way: 80% of the time is to use the '[mask]' as mark, and 10% of the time is to use a random word. Instead of the word mask, the original sentence remains unchanged 10% of the time. This method increases the model's access to information but does not allow the model to obtain total information. The NSP task is to select some sentence pairs A and B, 50% of the sentences B is the next sentence of A, and the remaining 50% of the sentences are randomly selected by the corpus so that the model can better learn the relevance between sentence and sentence.

In our model, we process the data to fulfill the input conditions of the Bert model. In the first step, the [CLS] mark is added at the beginning of the sentence and the [SEP] mark is added at the end. Then, shown in Fig. 2, after the two inputs consisting of the sum of three embeddings goes through the MLM process and NSP process in the same Bert model, the feature vectors about sentence is obtained. In the second step, Bert model fine-tune the feature vector from previous step for training the weights of our whole model to gain the optimal weight for our similarity tasks.

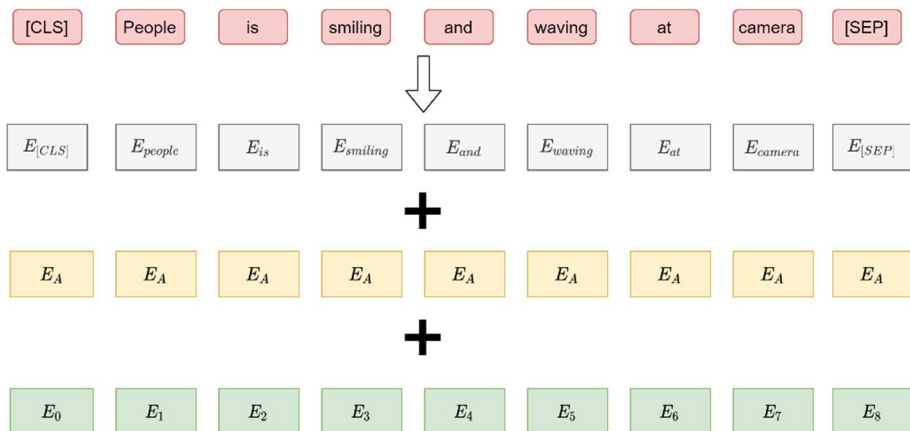


Fig. 2 Input composition of the Bert model

There are two types of Bert architecture with two versions of pre-trained weights, one is a 12 layers Bert-base and the other is a 24 layers Bert-large. The version of 12 layers is utilized for our task. We combine the Siamese network structure with the Bert model, which two sentences are input in the same Bert model and can be encoded respectively but reduce parameters and share weights. It is good for accelerating convergence than adopting two Bert models independently.

3.2 Siamese architecture model

Nowadays, the Siamese network model is mainly used for similarity measurement of images or texts. At first, it was applied to the field of computer vision, such as [11]. Later, researchers discovered that this method can also be applied to predict text similarity such as [28]. Compared with other methods, this method has achieved great success. The Siamese network is mainly composed of two branches. The two branches own the same structure and share parameters and weights. Because of the characteristics of sharing parameters and weights, the Siamese network model decreases the number of parameters during the training process, reduces the complexity of the model. Based on this feature, its double branch structure is very suitable for processing a given sentence pair and measuring the similarity of sentence pairs. In this way, we gain the representation vector of a pair of sentences in the same semantic space at the same time, which makes the comparison easier.

It is worth noting that the Siamese networks model is an architecture of the neural network, not a concrete network. Therefore, the CNN model or RNN model both can apply to its fulfillment. In this paper, we apply Bidirectional Long Short Term Memory (Bi-LSTM) model to build the Siamese architecture. Bi-LSTM network consists of a forward LSTM model and a backward LSTM model, which will better capture contextual semantic information and form bidirectional semantic dependence than a single-layer LSTM. LSTM network is a special RNN network to deal with RNN gradient problem. The core of LSTM is the cell state and an LSTM unit relies on three gates to control the cell state. These three gates are respectively called forget gate, input gate and output gate. The function of the forget gate is to determine what information the cell state needs to discard. The function of the input gate is to determine what new information is added to the cell state. The function of the output gate is to determine what the state information of the cell will be the final output. The final hidden state output of each LSTM unit can be gained by following Eqs. (1)–(6).

$$f_t = \sigma(W_f \cdot [h_{t-1}, e_t] + b_f) \quad (1)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, e_t] + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, e_t] + b_c) \quad (3)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (4)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, e_t] + b_o) \quad (5)$$

$$h_t = o_t * \tanh(C_t) \quad (6)$$

Where e_t is the input vector and h_t is the hidden state at time step t . W_f, W_i, W_o are the share weight matrices, b_f, b_i, b_o, b_c are biases vector. and σ is sigmoid function. C_t is the long memory cell devised to reduce the possibility of gradient disappearance and explosion gradient, which happen in the RNN model. i_t and o_t indicate the input gate and output gate separately. $\tilde{C}_t$ is the temporary state at time step t according to the forget gate f_t . $\tanh$ denote the activation function.

3.3 Attention mechanism

In our model, we will obtain a large amount of semantic feature information from the sentence yet different words have different contributions on the semantic expression of the entire sentence. We need the model to filter and judge which words representing semantic features are more important to the sentence and give them different weights in the sentence. Furthermore, we add Attention Mechanism to our model to solve this problem. In the beginning, the Attention Mechanism was inspired by human vision, when people see a picture, they tend to focus on a certain part of the picture and neglect other irrelevant information, which helps to improve their perception ability. Then researchers discovered that the text and relationship between word sequence can also be handled by Attention Mechanism. So Attention Mechanism extensively applies to a lot of NLP tasks [3, 23]. The Attention Mechanism is especially helpful to calculate semantic relevance among two feature representations in spite of their distance. Attention Mechanism can also better catch different distance context information for sentences [9, 14, 19]. Our motivation of using the Attention Mechanism for the semantic similarity task is to improve the proportion of important words that express more semantic information in each sentence for highlighting the semantic content of the sentence and compare the similarity between sentences.

In general, the role of the Attention mechanism enables the model to distinguish and selectively process more important information. In our model, the calculation of the attention mechanism is mainly divided into two steps: firstly, given two hidden states $H_x = [h_1, h_2, h_3, \dots, h_n]$ and $H_y = [h_1, h_2, h_3, \dots, h_m]$ as inputs. H represents the semantic feature vector of sentence, every h represents the semantic feature of each word in the sentence. We calculate the relevance of the semantic feature information of each word in H to the semantic representation of the sentence and gain the total attention distribution of each sentence in the two inputs x_t, y_t . Secondly, according to the attention distribution x_t, y_t , the attention mechanism would assign the weight of each word in each sentence and get the weight matrix of the sentence α_t, ϕ_t . The x_t, y_t and α_t, ϕ_t calculated by the following Eq. (7)~(10).

$$\alpha_t = \frac{e^{y_t}}{\sum_{i=1}^n e^{y_i}} \quad (7)$$

$$\phi_t = \frac{e^{y_t}}{\sum_{i=1}^m e^{y_i}} \quad (8)$$

$$x_t = \tanh(W_x H_x + b_x) \quad (9)$$

$$y_t = \tanh(W_y H_y + b_y) \quad (10)$$

Where W_x, W_y and b_x, b_y are the parameters of the attention mechanism for sentence. Finally, the output of the attention mechanism are $V_x = (v_1^\alpha, v_2^\alpha, v_3^\alpha, \dots, v_m^\alpha)$ and $V_y = (v_1^\phi, v_2^\phi, v_3^\phi, \dots, v_m^\phi)$, which represent higher level semantic expression. $v_t^\alpha = h_t \alpha_t$ and $v_t^\phi = h_t \phi_t$ respectively represent the semantic feature vector of each word in the sentence.

3.4 Evaluate similarity and loss function

When we obtain the high-level semantic expression of the two sentences from the neural network model of the Siamese architecture, we design two different methods of predicting similarity according to the type of dataset.

The first way obtains the spatial distance between two high-level semantic representations to represent the similarity according to the distance formula, and converts it into the predicted label value on the MSRP dataset and Quora dataset. Firstly, the distance between the two vectors is calculated by the distance measure function to generate a similarity score to evaluate the semantic similarity degree of the two sentences. Then we employ a sigmoid function to constrain the similarity score within the range [0,1]. We apply a threshold of 0.5 as score criterion. If the score > 0.5 , the predicted label is 1, otherwise the predicted label is 0. The sigmoid function formula is as follow (11):

$$f(x) = \frac{1}{1+e^{-x}} \quad (11)$$

In addition, this method is optimized during the training using the Contrastive loss function. In the traditional Siamese network architecture, contrastive loss is generally used as the loss function. This loss function is originally proposed by the paper of [11] and it is mainly used in dimensionality reduction, which means originally similar samples, after dimensionality reduction (feature extraction), in the feature space, the two samples are still similar; while the original dissimilar samples, after dimensionality reduction, in the feature space, the two samples are still not similar. Therefore, the contrastive loss function can effectively deal with the relationship of paired data in the Siamese neural network and adapt to train the model for extracting features. The loss function formula is as follow (12):

$$L = \frac{1}{2N} \sum_{n=1}^N yd^2 + (1-y)\max(\text{margin} - d, 0)^2 \quad (12)$$

Where $d = \|a_n - b_n\|^2$ represent the distance of two samples, y is the label of whether the two samples match. $y=1$ means the two samples are similar or matched, while $y=0$ means not matching. Margin is the threshold.

Another method is to calculate and concatenate high-level semantic expressions, and then use the softmax function to predict the similarity of sentences and convert them into corresponding labels. The high-level semantic expression of a sentence is essentially a high-level vector, so we perform a subtraction on the two high-level vectors to obtain the difference, and then concatenating the original two high-level vectors with the difference for obtaining a similarity matrix. Finally, the similarity matrix passes through two fully connected layers and utilize the softmax function to gain the prediction similarity distributions (13):

$$P(c) = \text{softmax}\left(W_s \left(V_x \oplus V_y \oplus |V_x - V_y| \right) + b_s \right) \quad (13)$$

where c is a similarity label with C classes, V_x and V_y are two high-level representation, $\oplus$ represents the concatenation operation, W_s and b_s are the parameters in the softmax function. Besides, this method is optimized during the training using the categorical-cross-entropy loss function. This loss function is used to evaluate the difference between the probability distribution obtained by the current training and the true distribution. It depicts the distance between the actual probability and the expected probability, that is, the smaller the value of cross entropy, the closer the two probability distributions are. The categorical-cross-entropy loss function formula is as follow (14):

$$L = - \sum_{i=1}^C y_i \log f_i(x) \quad (14)$$

where x is the input, y is the similarity label, C is the number of similarity label categories.

4 Experimental SETUP

In this section, we will introduce a series of experiments detail about proposed model. As part of section, we describe the datasets and evaluation metrics used in our experiments in section. Then, we present the training details and model parameter settings in section.

4.1 Dataset description

In our paper, we evaluate the effectiveness and performance of our proposed method on the three corresponding semantic similarity datasets including paraphrase identification task, text match task and natural language inference task.

MSRP The paraphrase identification task can be treated to determine whether two texts are paraphrase or not in line with their semantic similarity. Therefore, we perform experiments on the Microsoft Research Paraphrase dataset (MSRP) [8] to train and test model performance in textual semantic similarity task. MSRP dataset contains 5801 pairs of sentences and each label of sentence pair is 1 or 0 to indicate whether the two sentences are matched or not, which are extracted from news sources on the Web, and included manual annotations.

Quora The Quora Question Pairs (Quora) consists of 400,000 questions pairs and each question pair has a label which is 1 or 0 to determine whether the two questions contained in it express the same semantics. Approximately 37% of the Quora dataset are matched, and approximately 63% are unmatched.

SNLI The Stanford Natural Language Inference (SNLI) is proposed by [5]. This dataset contains 550,152 training samples, 10,000 validation samples, and 10,000 test samples. The three labels are applied to each pair of sentences according to their semantic similarity and semantic relationship, each pair of sentences being annotated with one of these labels.: entailment, contradiction, neutral.

4.2 Evaluation metrics

For evaluating the performance of our method and other methods on calculating similarity tasks, we use several evaluation indicators to compare. First, we introduce the matrix about TP, FP, FN, TN under Table 1.

Where TP(true positive) indicates the true label of the sample and the result of the final prediction is all positive, TN(true negative) is completely different from TP, which indicates that the true label of the sample and the result of the final prediction is negative. FP(false positive) indicates that the true label of the sample is negative but the result of the final prediction is positive, FN(false negative) indicates that the true label of the sample is positive, yet the result of the final prediction is negative. Then, our evaluation metrics are calculated from these values.

Accuracy Accuracy means the accuracy of the prediction result, the number of samples that are predicted correctly divided by the total number of samples. The formula (15) is as follow

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FN+FP} \quad (15)$$

Precision Precision means the prediction results which is the probability of predicted positive samples among the real positives. The formula (16) is as follow

$$\text{Precision} = \frac{TP}{TP+FP} \quad (16)$$

Recall Recall means the probability of actual positive cases are accurately predicted. The formula (17) is as follow

$$\text{Recall} = \frac{TP}{TP+FN} \quad (17)$$

F1-score F1-score means the harmonic average between precision and recall. The formula (18) is as follow

$$F1 = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \quad (18)$$

4.3 Distance metrics

The distance metric is used to measure the distance that an individual exists in space. The farther the distance, the greater the difference between individuals so we apply the distance between two semantic vectors to represent the similarity degree between two sentences. In our paper, we use three types of distance measure function, one is Euclidean distance, the second one is Manhattan distance and the other is Cosine

Table 1 Confusion matrix

Predict label/ True label	Positive	Negative
Positive	TP	FP
Negative	FN	TN

distance. Euclidean distance is the most ordinary distance benchmark, which calculates the absolute distance among points in a multidimensional space. Cosine distance utilizes the cosine value of the angle between two vectors in a multidimensional vector space as a amount of the disparateness among two individuals instead of directly measuring the distance of two points in the vector space. Manhattan distance was originally calculated in the standard coordinate system to calculate the actual distance between two points. But in high-dimensional space, Manhattan distance can also be used to calculate the sum of the absolute difference between two vectors to represent their actual distance, rather than the geometric distance derived from the Euclidean distance to represent the actual difference. We conducted experiments to demonstrate the performance and differences of three methods and choose the best outcome as our model. In our experimental results section, we will compare the final performance of these three methods of distance measure function on the dataset, and take the best result as the best performance of our model. The Euclidean distance formula, Cosine distance formula and Manhattan distance function are as follows (19)–(21):

$$\text{Euclidean}_{\text{dis}(X,Y)} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (19)$$

$$\text{Cosine}_{\text{dis}(A,B)} = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (20)$$

$$\text{Manhattan}_{\text{dis}(X,Y)} = \sum_{i=1}^n |x_i - y_i| \quad (21)$$

4.4 Training detail

In this paper, we utilized the 12 layers Bert-based version for our task. Two sentences are input into the same Bert model and can be encoded separately but share weights and reduce parameters. It is good for accelerating convergence than using two Bert models directly. In the last layer of the Bert model, we obtain the sequence of hidden-states. Then we use two layers Bidirectional LSTM model to construct the Siamese network architecture. The dimension of the output hidden state is set to 256 and 128 in the first layer and the other layer respectively. In order to avoid overfitting, dropout with the rate of 0.4 is applied to every output layer of the Bidirectional LSTM model. We employ Adam Optimizer as our optimizer with initial learning rate of 1e-4 in the training process. The Adam optimizer uses the calculation of the gradient, dynamically adjusts the learning rate of each parameter, and constrains the learning rate, so that each learning rate has a certain range, and the parameters will be relatively stable. We performed batch normalization regularization to the semantic feature vectors for each training mini-batch. Moreover, we applied early-stopping technique during our training process to avoid overfitting and achieve the best model performance. The batch-size is set to 32 on the MSRP dataset and Quora dataset. On the SNLI dataset, we set the batch-size to 16.

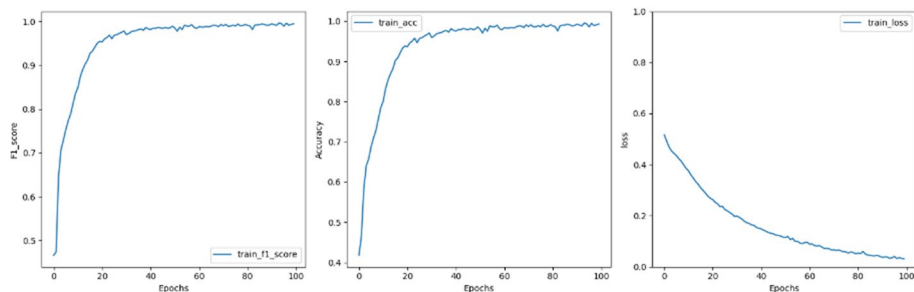


Fig. 3 F1, accuracy, loss curves during training process on the MSRP dataset

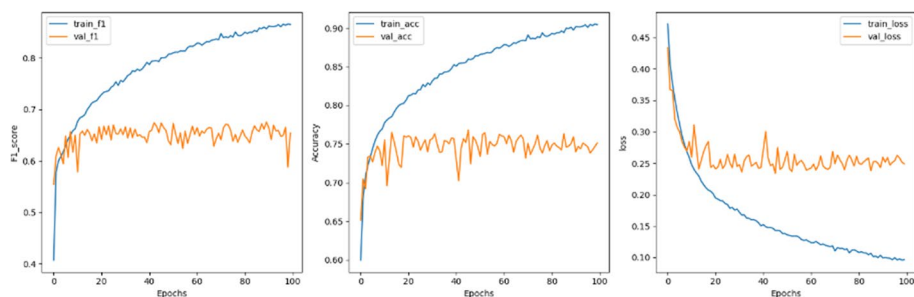


Fig. 4 F1, accuracy, loss curves during training process on the Quora dataset

5 Experimental results and analyses

In this section, we will introduce a series of experimental results to test the performance comparison of our model and other models under different dataset related to semantic similarity task. Through the experimental results, we prove the effect and generality of our model.

5.1 Training process

Figures 3, 4 and 5 respectively show the change curves of ACC, F1, and loss during the model training process on the MSRP dataset, Quora dataset and SNLI dataset. Since MSRP is a small dataset and no verification data set is set, so there is no indicator curve for the verification set in Fig. 4. From Figs. 4 and 5, we can see that as the number of training epochs increases, all curves first show an upward trend, and finally oscillate back and forth, and finally stabilize. From comparing the validation set and the training set, we found that although the accuracy of the training set has been improving and the training loss value has decreased as the training epochs have increased, the accuracy of the validation set has fluctuated, and the loss value has increased, indicating the model is overfitting. Therefore, during the training process, we use the early stopping technique, which is to stop the model training after the accuracy of the validation set does not increase or the loss value does not decrease.

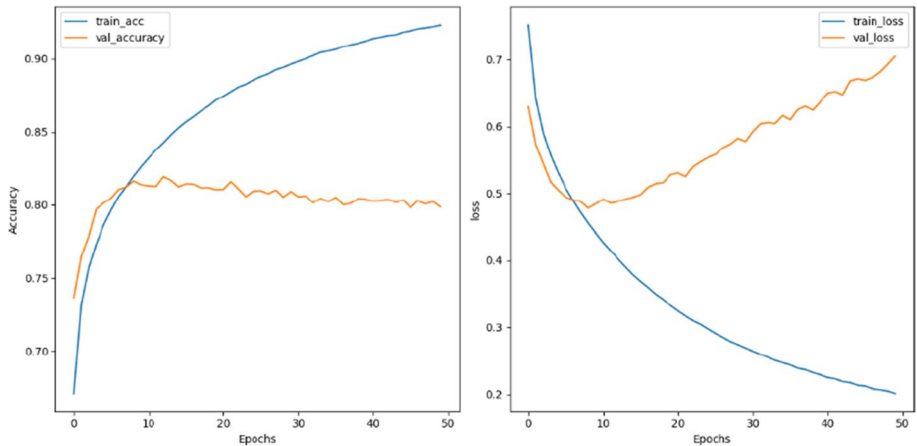


Fig. 5 Accuracy, loss curves during training process on the SNLI dataset

5.2 Baseline

We compare our model with a number of baseline models on the Quora dataset, MSRP dataset and SNLI dataset. The baseline model for comparison is part of the traditional model like the BLSTM model, and part of the recent studies in textual similarity tasks.

The baseline models and proposed algorithms for comparison on MSRP dataset:

- BLSTM model used a traditional LSTM network to encode sentences.
- Attention-BLSTM model used a LSTM network with attention to encode sentences.
- CA-RNN model incorporated the contextual information of the aligned words in a sentence pair for modeling sentence similarity [6].
- Multi-scale MaLSTM model proposed a Multi-scale Siamese LSTM network with hierarchical sentence factorization to assess semantic similarity between sentences [21].
- Multi-scale HCTI model proposed Multi-scale CNN system with hierarchical sentence factorization for textual semantic similarity [21].
- DPIM-ISS model introduced the linguistic characteristic demonstrated in syntactic attributions to generate more explicit construction and encoded the semantic representation of sentence on various syntactic structure [17].

The baseline models and proposed algorithms for comparison on Quora dataset:

- Support vector machine (SVM) mean the model utilizes several hand-crafted features in a Support Vector Machine (SVM) model [15].
- Random Forest used an ensemble of decision trees to train data and average the result of all trees to make prediction [39].
- K-Nearest Neighbor utilized the average of k-nearest data points to predict the testing data value [39].
- Logistic Regression is used for predicting an outcome from independent variables to fall into any of the available class [39].
- Siamese-CNN used CNN model to construct the Siamese architecture model for the textual semantic similarity task [42].

- BLSTM model used a traditional LSTM network to encode sentences.
- AugDeepParaphrase utilized a CNN model and a LSTM model on coarse-grained sentence modeling, combined with a specific fine-grained word-level similarity matching model [1].
- Multi-Perspective-CNN applied multiple perspective cosine matching function based on CNN network [42].
- Multi-Perspective-LSTM applied multiple perspective cosine matching function based on LSTM network [42].

The baseline models and proposed algorithms for comparison SNLI dataset:

- LSTM encoder is a traditional LSTM model which utilized LSTM network encode sentences [5].
- SPINN constructed tree-structured LSTM to encode sentence [4].
- Tree-CNN model constructed tree-structured CNN to encode sentence [26].
- Attention-LSTM model utilized LSTM architecture with attention mechanism to gain the semantic representation of each sentence [33].

5.3 Experimental results and analysis

The performance of our Siamese Bert model with attention mechanism is compared to the baseline model's performance on the MSRP dataset, Quora dataset, and SNLI dataset. Table 2 summarizes the experimental results based on the MSRP dataset, Table 3 summarizes the results based on the Quora dataset, and Table 4 summarizes the results based on the SNLI dataset. Our model performs well on these three representative datasets for semantic similarity-related tasks.

Table 2 displays the evaluation metrics like accuracy and F1-score of other models on the MSRP dataset. Our proposed model performs better than existing approaches for semantic similarity tasks with an accuracy value of 77.77% and an F1-score of 84.9, based on the table above.

As we can see in Table 3, we apply machine learning algorithms like SVM, Random forest, KNN and Logistic Regression to semantic similarity task on the Quora dataset. But our model outperforms the other algorithms in evaluating the similarity-score and prediction of semantic similarity within the sentences, which the accuracy of 85.15% is greater than SVM 75.32%, Random Forest 72.3%, KNN 71.9%, Logistic Regression 67.1%, while the F1-score of 74.64 is higher than SVM 60.89, Random Forest 73.0,

Table 2 Performance of models on the MSRP dataset

Model	Accuracy(%)	F1-score(%)	Reported from
BiLSTM	73.60	81.80	[38]
Attention-BiLSTM	75.40	82.70	[38]
CA-RNN	77.30	84.00	[6]
Multi-scale MaLSTM	74.09	82.18	[21]
Multi-scale HCTI	74.03	81.76	[21]
DPIM-ISS	73.57	83.55	[17]
Our model	77.77	84.49	

Table 3 Performance of models on the Quora dataset

Model	Accuracy(%)	F1-score	Reported from
SVM	75.32	60.89	[43]
Random Forest	72.3	73.0	[39]
KNN	71.9	72.0	[39]
Logistic Regression	67.1	69.0	[39]
BLSTM	75.53	75.53	[43]
Siamese-CNN	79.60	-	[42]
AugDeepParaphrase	77.70	-	[25]
Multi-Perspective-CNN	81.38	-	[42]
Multi-Perspective-LSTM	83.21	-	[42]
Our model	85.15	78.65	

Table 4 Performance of models on the SNLI dataset

Model	Accuracy(%)	Reported from
LSTM encoder	78.2	[5]
SPINN	80.6	[4]
Tree-CNN	82.1	[26]
Attention-LSTM	83.5	[33]
Our model	83.82	

KNN 72.0, Logistic Regression 69.0. In addition, the accuracy and F1-score of our presented model are better than the traditional neural network models and the previous approaches on semantic similarity task.

The above Table 4 illustrates the accuracy rate value representation comparison of proposed model with other methods such as LSTM model, SPINN model, Tree-CNN model and Attention-LSTM model on the SNLI dataset. Our proposed framework of Siamese Bert architecture with attention mechanism generates the accuracy value of 83.82%, which is higher than the other model. The experimental results show that our proposed model also has high performance on natural language inference related to semantic similarity tasks, which indicates that our model has some generality on semantic similarity tasks.

We have also made different changes to our Siamese network structure while doing our research. Our Siamese network model has two branches owning the same structure and sharing parameters and weights with each other. Furthermore, we change the two branches of shared parameters in the Siamese network structure to two independent branches. Such a network can be called Pseudo-Siamese network whose training parameters are nearly double the training parameters of the Siamese network. The Fig. 6 illustrates the performance comparison of Pseudo-Siamese network and Siamese network on different datasets. The comparisons evaluate that the Siamese network exhibits higher rate of accuracy ranges in determining semantic similarity NLP-tasks than Pseudo-Siamese network. Table 5 shows the performance indicators accuracy and F1-score of the Siamese network on the MSRP dataset, on the Quora dataset and on the SNLI dataset, which are better than Pseudo-Siamese network regarding the prediction of the textual semantic similarity and reduces the parameters. Therefore, the Siamese network is more suitable for the semantic similarity task of sentence pairs than the Pseudo-Siamese network.

Fig. 6 Performance comparison of Pseudo-Siamese network and Siamese network

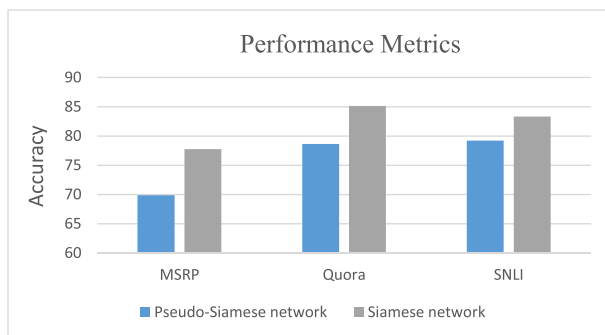
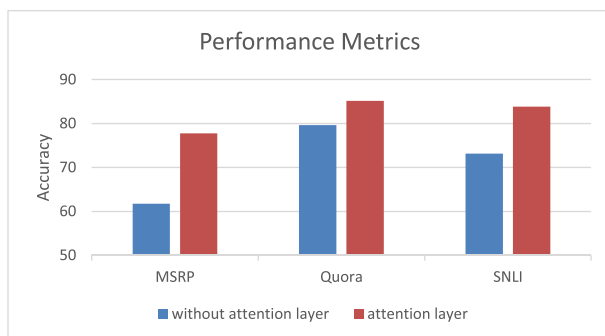


Table 5 Comparison of different architecture model on MSRP Dataset and Quora Dataset

Dataset	Model	Accuracy(%)	F1-score
MSRP	Pseudo-Siamese network	69.87	80.01
	Siamese network	77.77	84.49
Quora	Pseudo-Siamese network	78.64	74.32
	Siamese network	85.15	78.65
SNLI	Pseudo-Siamese network	79.21	-
	Siamese network	83.82	-

Fig. 7 Performance comparison of without attention layer and attention layer

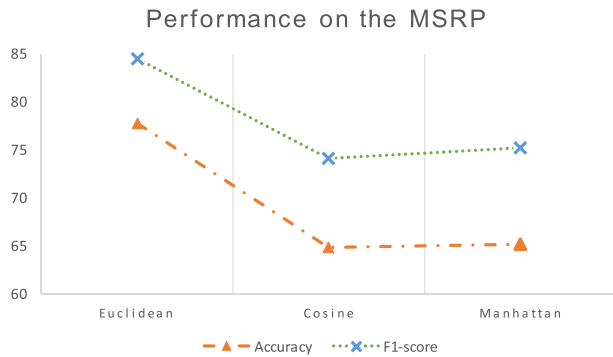
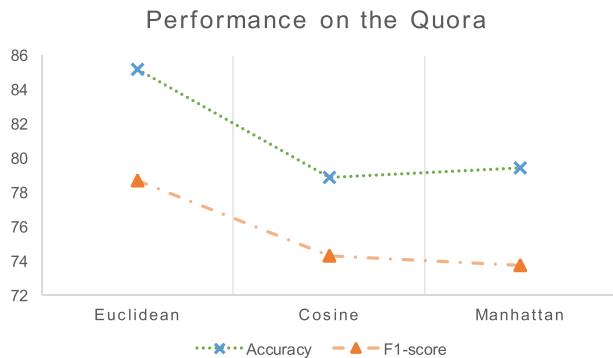


The proposed framework relying in different datasets is made compared with the accuracy rate of without attention mechanism model in this Fig. 7. Obviously, the outcomes of the model with the attention mechanism is better than that without the attention mechanism on the different datasets. In Table 6, the model with an attention mechanism outperforms one without it in accuracy value and F1-score value on the datasets related to semantic similarity. We can conclude that the attention mechanism gives the weight of each word in the sentence to gain a higher level of semantic representation and is helpful to improve the performance of the model.

Besides, we also test the performance of different methods of measuring distance on the performance of our model such as Euclidean distance, Cosine distance and Manhattan distance. The comparison of the experimental results is in Figs. 8 and 9. Using Cosine distance as the method of measuring the distance of two vector texts, the accuracy and f1-score on

Table 6 Different methods on the MSRP Dataset and Quora Dataset

Dataset	Model structure	Accuracy(%)	F1-score
MSRP	without attention	61.73	62.26
	attention	77.77	84.56
Quora	without attention	79.64	74.33
	attention	85.15	78.65
SNLI	without attention	73.16	-
	attention	83.82	-

Fig. 8 Different methods of measuring the distance on the MSRP dataset**Fig. 9** Different methods of measuring the distance on the Quora dataset

the MSRP dataset and Quora dataset are respectively 64.86, 74.18 and 78.88, 74.28. Using Manhattan distance as the method of measuring the distance of two vector texts, the accuracy and f1-score on the MSRP dataset and Quora dataset are respectively 65.15, 75.28 and 79.44, 73.75. The accuracy and f1-score of the method using Euclidean distance on the MSRP dataset and Quora dataset are respectively 77.77, 84.56 and 85.15, 78.65. Obviously, using Euclidean distance as a method of measuring distance has better performance than using cosine distance or Manhattan distance in our network model.

6 Conclusion and future work

Textual semantic similarity prediction is one of the most significant tasks in the field of NLP. In our paper, we have proposed the Siamese Bert Architecture Model with Attention Mechanism to apply to this task. The Siamese network structure is combined with bidirectional LSTM units that capture the features of the shared weight and use Bert to extract text features, and the attention mechanism is used to improve the semantic representation of vector texts. Finally, we use the different method to predict the semantic similarity between two vector texts. Moreover, we conduct experiments on the three public datasets related to different semantic similarity prediction tasks. Based on experimental results on three datasets, the presented model for semantic similarity prediction performs well compared to some traditional and recent approaches, proving its generalizability. Our future work will focus on comparing the two texts with more attentional interaction in order to optimize our model and add noisy datasets to enhance adversarial training. Additionally, we wish to add a domain adaptation technique to increase generality in a variety of domains.

Acknowledgements This research is supported by Key-Area Research and Development Program of Guangdong Province under Grant 2019B010153002, Key Program of NSFC-Guangdong Joint Funds under Grant U1701262 and U1801263, National Natural Science Foundation of China under Grant 62002071 and Guangdong Provincial Key Laboratory of Cyber-Physical System under Grant 2020B1212060069.

Data availability The data generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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